

# Constant Magnetic Flow on the Hyperbolic Plane: Orbit Trichotomy, Critical Horocycles, and Exact Periods

Route-A source-local certificate HCS-C289

2 September 2026

## Abstract

On the complete simply connected surface of curvature  $-\kappa^2$  we solve  $D_t\dot{\gamma} = bJ\dot{\gamma}$  at fixed speed  $v$ . For  $v > 0$ , every trajectory is a circle, horocycle, hypercycle, or geodesic, according as  $|b|$  is greater than, equal to, or less than  $\kappa v$  (with  $b = 0$  separated). Only the circle chamber closes, with primitive period  $2\pi/\sqrt{b^2 - \kappa^2 v^2}$ . At equality the Lorentz-frame generator is nonzero nilpotent, and the orbit is a nonclosed horocycle. We prove the classification globally from a raw  $3 \times 3$  moving-frame equation.

## 1 Model, conventions, and result

Let  $\mathbb{H}_\kappa^2$  be the oriented complete simply connected Riemannian surface with Gaussian curvature  $-\kappa^2$ , where  $\kappa > 0$ , and let  $J$  denote positive rotation through  $\pi/2$ . We study

$$D_t\dot{\gamma} = bJ\dot{\gamma}, \quad |\dot{\gamma}| = v, \quad (1)$$

for constant  $b \in \mathbb{R}$ . Metric compatibility makes the speed constant; we nevertheless retain  $v$  because replacing  $b$  by the geodesic curvature is a frequent source of a missing factor. Time in every period statement is the physical parameter  $t$  in (1).

The orbit geometry is classical. Comtet treats the hyperbolic constant-field problem and its Landau quantization [1]; Adachi treats magnetic trajectories on constant-holomorphic-curvature spaces [2]. No literature originality is claimed for the classification below. Our purpose is a convention-complete derivation joined to independently replayable evidence.

**Theorem 1** (Complete orbit atlas). *Assume  $\kappa > 0$  and  $v > 0$ . Every solution of (1) is complete. Up to an orientation-preserving isometry of  $\mathbb{H}_\kappa^2$  and a time translation:*

1. *if  $|b| > \kappa v$ , it is a hyperbolic circle and its primitive period is*

$$T = \frac{2\pi}{\sqrt{b^2 - \kappa^2 v^2}}; \quad (2)$$

2. *if  $|b| = \kappa v$ , it is a nonclosed horocycle;*
3. *if  $0 < |b| < \kappa v$ , it is an unbounded hypercycle;*
4. *if  $b = 0$ , it is a geodesic.*

*The sign of  $b$  reverses the orientation but not the unoriented curve or its period. At the critical face the ambient generator has  $A^3 = 0$  and  $A^2 \neq 0$ .*

**Boundary convention.** If  $v = 0$ , the curve is stationary and the unit-tangent proof below is not invoked. The flat case  $\kappa = 0$  is a limit, not a point in the theorem:  $b \neq 0$  gives a Euclidean circle of period  $2\pi/|b|$ , whereas  $b = 0$  gives a straight line.

## 2 The constant Frenet system

For  $v > 0$  put  $T = \dot{\gamma}/v$  and  $N = JT$ . Equation (1) becomes

$$D_t T = bN, \quad D_t N = -bT, \quad (3)$$

so the signed geodesic curvature with respect to arclength is  $k_g = b/v$ . Use the hyperboloid

$$\mathbb{H}_\kappa^2 = \{X \in \mathbb{R}^{2,1} : \langle X, X \rangle_L = -\kappa^{-2}, X_0 > 0\}, \quad \eta = \text{diag}(-1, 1, 1).$$

Set  $e_0 = \kappa X$ . The ambient derivative of a tangent field  $Y$  along  $X$  is

$$\frac{dY}{dt} = D_t Y + \kappa^2 \langle \dot{X}, Y \rangle X, \quad (4)$$

or, in particular,  $\dot{T} = \kappa v e_0 + bN$ . With the signs in (3), let  $F = (e_0, T, N)$  explicitly denote the matrix whose columns are those three vectors. It satisfies the right-action equation  $\dot{F} = FA$ , with

$$A = \begin{pmatrix} 0 & \kappa v & 0 \\ \kappa v & 0 & -b \\ 0 & b & 0 \end{pmatrix}, \quad A^\top \eta + \eta A = 0. \quad (5)$$

Changing from row to column frames transposes the displayed ODE but changes neither the characteristic polynomial nor any conclusion.

**Lemma 1** (Cubic identity). *For (5),*

$$\det(\lambda I - A) = \lambda(\lambda^2 + b^2 - \kappa^2 v^2), \quad A^3 = (\kappa^2 v^2 - b^2)A. \quad (6)$$

At  $|b| = \kappa v$ ,  $A^2 \neq 0$  and  $A^3 = 0$ .

*Proof.* Direct expansion of the three-by-three determinant proves the first identity; Cayley–Hamilton proves the second. Substitution  $b = \pm \kappa v$  leaves, for example, the  $(1, 1)$  entry of  $A^2$  equal to  $\kappa^2 v^2 > 0$ , so the critical generator is nonzero of nilpotency index three.  $\square$

## 3 Proof of the classification

*Proof of Theorem 1.* The frame equation has the global solution  $F(t) = F(0)e^{tA}$ . Thus every initial oriented frame gives a solution for all  $t$ , and transitivity of  $\text{SO}^+(2, 1)$  on such frames shows that the analysis exhausts all magnetic initial data.

If  $\delta = b^2 - \kappa^2 v^2 > 0$ , (6) gives eigenvalues  $0, \pm i\sqrt{\delta}$ . The one-parameter Lorentz subgroup is elliptic and fixes a timelike axis. Intersecting the hyperboloid with the affine plane at fixed Lorentz distance from that axis gives a hyperbolic circle. On the two-dimensional rotating block, put  $e_0 = (1, 0, 0)^\top$  in frame coordinates. The cubic identity gives

$$e^{tA} e_0 = e_0 + \frac{\sin \theta}{\sqrt{\delta}} A e_0 + \frac{1 - \cos \theta}{\delta} A^2 e_0, \quad \theta = t\sqrt{\delta}.$$

Here  $(Ae_0)_T = \kappa v \neq 0$ ,  $(A^2 e_0)_T = 0$ , and  $(A^2 e_0)_{e_0} = \kappa^2 v^2 \neq 0$ . A position return first forces  $\sin \theta = 0$  from the tangent component and then forces  $\cos \theta = 1$  from the  $e_0$  component. Thus the base point returns exactly for  $\theta \in 2\pi\mathbb{Z}$ , and (2) is primitive.

If  $\delta = 0$ , Lemma 1 gives a nontrivial parabolic subgroup. Its hyperboloid orbits are horocycles. More specifically,  $e^{tA} e_0 = e_0 + tAe_0 + t^2 A^2 e_0/2$ , whose tangent component is  $\kappa vt$ . It is nonzero for every  $t \neq 0$ , so the base point itself never returns; the critical horocycle is not closed.

If  $\delta < 0$ , the nonzero eigenvalues are real and the subgroup is hyperbolic. Its nongeodesic hyperboloid orbits are hypercycles. When  $b = 0$ , the normal is parallel and the curve is the invariant geodesic itself. The sign change  $b \mapsto -b$  is conjugated by reversing  $N$ , so it reverses orientation and preserves the remaining data.  $\square$

## References

- [1] A. Comtet, *On the Landau Levels on the Hyperbolic Plane*, Annals of Physics 173 (1987), 185–209. [doi:10.1016/0003-4916\(87\)90098-4](https://doi.org/10.1016/0003-4916(87)90098-4).
- [2] T. Adachi, *Kaehler Magnetic Flows for a Manifold of Constant Holomorphic Sectional Curvature*, Tokyo Journal of Mathematics 18 (1995), 473–483. [doi:10.3836/tjm/1270043477](https://doi.org/10.3836/tjm/1270043477).